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An efficient pipelined architecture for real-valued Fast Fourier Transform

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ABSTRACT
Real-valued Fast Fourier Transform (FFT) plays an important role in today's digital world because of the fact that most of the signals contain real values. The FFT computation of real signals using conventional techniques requires more hardware space with high power consumption, which is the most important task for a researcher while designing VLSI architectures. This can be eradicated by clearly analysing the symmetric property of the real-valued signals. In this paper, we have adopted the symmetric property and designed an efficient pipelined architecture for 16-point DIF FFT. The pipeline scheme reduce the processing time at the cost of some registers and in order to contribute efficiently for power reduction we have modified the complex multiplier with reduced internal real multipliers which are in turn replaced by an modified canonic signed digit multiplier (CSDM) with resource-sharing technique. The complete module is synthesised and simulated using Xilinx ISE 14.1 with the target device is Virtex-5 xc5vlx110T. The experimental results verify that our implemented design is more efficient in terms of speed, area and power when comparing with similar works.

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KEYWORDS
Real Fast Fourier Transform (RFFT); Fast Fourier Transform (FFT); complex multiplier; canonic signed digit multiplier (CSDM); Pipelining

1. Introduction

Fast Fourier Transform (FFT) and its opposite [inverse FFT (IFFT)] are most generally utilised in calculation as a part of advanced sign preparing applications (Oppenheim, Schafer, & Buck, 1998; Teymourzadeh, Abigo, & Hoong, 2014). Ordinarily, FFT/IFFT works over complex numbers and delivers complex numbers. A few plans had been introduced for calculation of FFT and IFFT of complex numbers [complex FFT (CFFT)/complex IFFT (CIFFT) (He & Torkelson, 1998; Lin & Lee, 2007; Lin, Liu, & Lee, 2005). The enthusiasm for calculation of quick Fourier change (FFT) for genuine esteemed signs (RFFT) has been expanding, subsequent to about the greater part of the physical signs, which is genuine and set up quick Fourier change of Hermitian-symmetric signs, alluded to as genuine set up quick Fourier change (RIFFT) (Oppenheim et al., 1998). In this manner, about portion of the calculations are repetitive. This property can be abused to lessen the equipment multifaceted nature. Pipelined equipment design is broadly utilised for calculation of FFT, in light of the fact that it offers high throughput, low inactivity, low range and power utilisation (Salehi et al., 2013).

The RFFT has real impact in various fields, for example, correspondence frameworks, biomedical applications, sensors and radar signal handling. The primary thought for utilising genuine esteemed FFT is to locate the middle symmetries that happen in the calculation while changing

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Efficient Audio Noise Reduction System Using Butterworth Chebyshev and Elliptical filter

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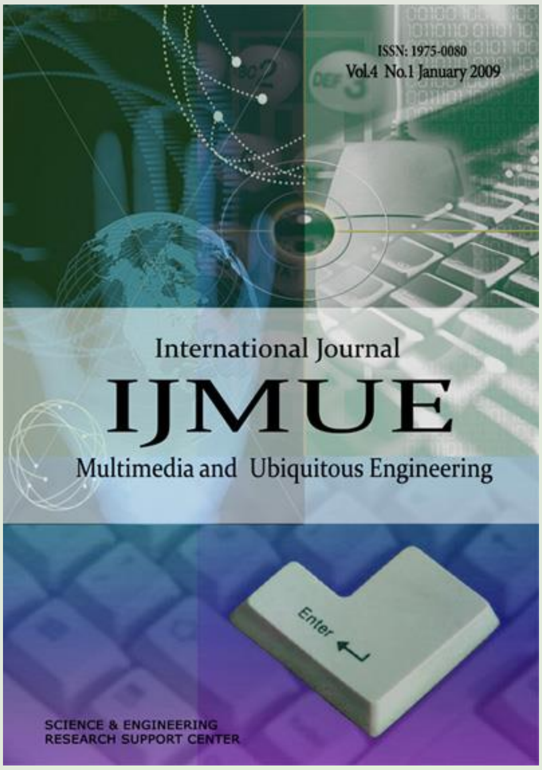

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Efficient Audio Noise Reduction System Using Butterworth Chebyshev and Elliptical filter

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Abstract

Digital signal processing is widely used to manipulate, modify, enhance or filter signals such as speech, audio, image and telecommunication signal. Signals can be processed in the analog domain, but the digital domain offers high speed, better accuracy, greater flexibility increased storing capabilities and simple implementation. The audio signal noise reduction has become a fundamental area of research for many real world applications. There are two forms of filters one is fixed, and another one is tunable. Fixed filters measure those within which passband frequencies and stop band frequencies, square measure mounted whereas just in case of tunable filters, passband and stop band frequencies measured variable. These frequencies will be modified consistent with the need of the applications. Tunable digital filters are widely used in medical electronics, digital audio instrumentation, telecommunications and control systems. It is often the essential requirement for removal of noise from the audio signal.

Keywords*: Fixed Filters, Tunable Filters, Audio Noise Reduction System*

1. Introduction

It is vital for engineers of inserting frameworks, processors, and tools targeting video applications. The audio noise reduction system is the system that's accustomed take away the noise from the audio signal. Audio noise reduction systems are classified into two basic processes. Coming to the primary process is that the complementary sort that involves compressing the audio signal in a well-defined manner before it's recorded. The second method is that the single-ended or non-complementary sort that utilizes techniques to cut back the noise level already placed within the supply material—in essence a playback solely noise reduction system. This method is employed by the LM1894 integrated circuit, is designed for the reduction of hearable noise in nearly any audio source. Noise reduction is the method of removing noise in the signal. All analog or digital recording devices have traits that build them vulnerable to noise [15].

In device's mechanism or processing algorithms having noise. That will be white or random noise without coherence. There can be Active noise control (ANC), also called noise cancellation. Active noise reduction (ANR) may be a technique to reduce unwanted and unprocessed sound by the addition of a second sound specifically designed to cancel the first. The sound may be a pressure wave, or we may say the sound is that the analog signals that are processed consistent with their frequency, that is consisting of a rarefaction phase and compression phase. A noise-cancelling speaker is emitting a sound wave with an equivalent amplitude, however, with inverted phase (also called antiphase) to the original sound signal [9]. The waves are mixed to make the new wave, in an exceedingly process known as interference, and effectively cancel one another out an impression that is named phase cancellation.

Today's advanced active noise control is achieved usually through the employment of analog circuits or digital signal process. An associate adaptive algorithm, the square

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Complex-multiplier implementation for pipelined FFTs in FPGAs

2
Author(s)

M. Aravind Kumar ; K. Manjunatha Chari

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Abstract:
Different approaches for implementing a complex multiplier in pipelined FFT are considered and implemented to find an efficient one in this paper. The design is implemented in VHDL and design is synthesized on FPGA to know the performance. The design is implemented with a focus of reducing the resources used. Some approaches resulted in the reduced number of DSP blocks and others resulted in reduced number of LUTs. Analysis of Synthesis results is performed for different widths (bit lengths) of complex multiplier approaches.

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INSPEC: Controlled Indexing
table lookup, fast Fourier transforms, field programmable gate arrays, hardware description languages, multiplying circuits, signal processing

INSPEC: Non-Controlled Indexing
LUT, complex-multiplier implementation, pipelined FFT, FPGA, VHDL, DSP blocks

Author Keywords
utilization and twiddle factors, VHDL, FFT, FPGAs, complex multiplier, LUT, DSP block

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Related Articles

An Efficient VLSI Architecture Decimation-in-Frequency Fast Fourier Transform Algorithm for Real-Valued Data

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Abstract **- The Decimation-in-frequency (DIF) fast Fourier Transform (FFT) for real-valued applications, like speech/image/video processing, bio medical signal processing, and time-series analysis, etc. Besides the DIF FFT butterfly involves less computation time than DIT counterpart. In this paper, we presents an efficient architecture for radix-2 DIF real real-valued FFT (RFFT).We presents here the necessary mathematical formulation for removing the redundancies in the radix-2 DIF RFFT, and present a formulation to regularize its flow graph to facilitate folded computation with a simple control unit. We propose here a register based storage design which involves significantly less area at the cost of higher latency compared with the conventional RAM-based storage. The address generation for folded in-place DIF RFFT computation with register based storage is challenging since both read and write operations are performed in the same clock cycle at different locations. Therefore we presents here a simple formulation of address generation for the proposed radix-2 DIF RFFT structure.**

Keywords: DIF-FFT, RFFT, FPGA

I. Introduction

Fast Fourier Transform (FFT) is a commonly used technique for the computation of Discrete Fourier Transform (DFT). DFT calculations are required in the fields like filtering, spectral analysis, etc. to calculate the frequency spectrum or to identify a system's frequency response from its impulse response and vice versa. FFT is used in digital video broadcasting and OFDM systems. Much research has been carried out to design pipelined architectures for computation of FFT. The basic one is Radix-2 FFT. Based on the radix-2 FFT approach many algorithms have been developed which includes radix-4, split-radix, etc. A popularly known algorithm is Cooley-Tukey radix-2 FFT.Radix-2 Multipath Delay Commutator (R2MDC) is a classical approach for pipelined implementation of FFT architecture. Radix-2 Single-path Delay Feedback (R2SDF) is another approach with reduced memory obtained by a standard usage of storage buffer in R2MDC [1][2].

Most of the algorithms require hardware complexity, and there is no complete hardware utilization. The basic aspects like high throughput and low power consumption are required to speed and power requirements, keeping the hardware overhead to a minimum. This paper presents a technique to design the architecture from the FFT flow graph.

A. Related Work:

Previously, in the past various algorithms for computation of RFFT is presented. But they did not have proper regular geometry. This is very important for deriving pipelined architecture. Firstly pipelined architecture for the real-valued signal was designed. But it is restricted to only radix-2 and 4 parallel RFFT architecture. Also, it has only real datapaths. To derive FFT architecture for real and complex inputs a design is presented. But in this architecture, even after removing redundant operations, it still calculates these samples. Again, there was no full hardware utilization of architecture derived in this paper. In pipelined RFFT architecture is derived but it has more hardware complexity than our proposed architecture as it has some adders, multipliers and delays than radix-2^3algorithm.

1) FFT Architecture via Folding Transformation:

Folding Transformation and Register minimization techniques are used to derive several FFT architectures. The whole process is explained with the help of 8-point radix-2 DIF FFT which can be extended to different radices. The flow graph of8-point radix-2 DIF FFT is illustrated in below

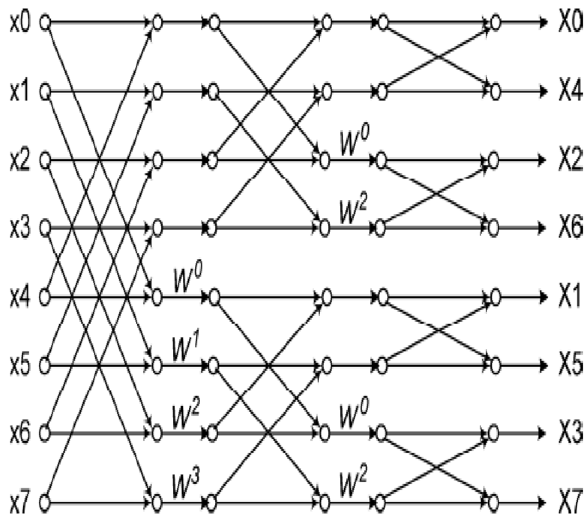

Fig. 1. Flow graph of a radix-2 8-point DIF FFT